

■ Organic & Supramolecular Chemistry

The Synthesis of Tetrasubstituted Cycloalkenes Bearing π -Conjugated Substituents and Their Optical PropertiesMarek Čubiňák,^[a] John Bignon,^[b] Pavel Galář,^[b] Lukáš Ondič,^[b] and Tomáš Tobrman^{*[a]}

We report the development of a new methodology for the preparation of cyclic tetrasubstituted alkenes with π -conjugated substituents. The synthesis of the target compounds was accomplished through the regioselective cross-coupling reaction of diphenyl bromocyclohexenyl phosphate with organo-

zinc reagents. The procedure succeeded in the preparation of alkenes with two pyrene moieties. The absorption and photoluminescence spectra of the prepared compounds were measured.

Introduction

A cyclic or acyclic tetrasubstituted double bond represents a valuable motif in a wide variety of substances. Cephalosporin P₁,^[1] fusidic acid,^[2] and tamoxifen^[3] are all examples of compounds that have either been isolated from natural resources or are used for their biological properties. Another example of a tetrasubstituted alkene is tetraphenyl ethylene (TPE), which is considered a cornerstone in the field of materials chemistry.^[4] Thus, hexaphenyl,^[5] meta-terphenyl,^[6] [2.2] paracyclophane,^[7] 9,10-diazaanthracene,^[8] cyclopentadiene,^[9] and naphthalene^[10] have all been decorated with TPE units to determine the physical properties of the prepared compounds. Such efforts have resulted in, for example, the design of novel blue tandem organic light-emitting diodes^[11] or a fluorescent probe for the detection of hypochlorite.^[12] A TPE unit is also a suitable candidate for the synthesis of polycyclic aromatic hydrocarbons (PAHs) by means of 6π -electrocyclization.^[13]

Yet, pyrene is a representative example of a simple PAH that also plays an important role in materials chemistry. The applications of substituted pyrenes include the preparation of fluorometric and colorimetric sensors,^[14] the formation of metal-organic frameworks,^[15] the construction of probes for studying protein conformation and conformational changes,^[16] and the development of supramolecular chemical sensors.^[17] The significant properties of both TPE and substituted pyrenes have led to the design and attempted preparation of a wide

range of tetrasubstituted alkenes featuring a tetrasubstituted double bond and a pyrene moiety. Although a wide variety of procedures are available for the preparation of substituted pyrenes^[18] and tetrasubstituted alkenes,^[19] the preparation of alkenes with four, three, two, or one pyrene unit(s) can only be achieved using a few procedures. For instance, alkenes bearing four 4,5,9,10-tetrahydropyren-2-yl units can only be synthesized using the McMurry reaction,^[20] whereas alkenes bearing two pyrenyl units directly attached to the double bond can be synthesized by means of the McMurry reaction,^[21] the Wittig reaction,^[22] Horner-Wadsworth-Emmons (HWE) modification,^[23] metathesis,^[24] carbonyl group condensation,^[25] and the rhodium-catalyzed deoxygenation-Suzuki reaction sequence.^[26] A similar trend can also be observed in relation to alkenes with two pyrene units that are connected to the double bond via a spacer. Such substances can be efficiently synthesized through the formation of a substituted double-bond moiety via the McMurry reaction^[27] or cross-metathesis^[28] followed by the Suzuki reaction. The widest range of procedures for the preparation of pyrenyl alkenes is available for alkenes with one pyrene moiety. In addition to the above-mentioned procedures, dehydration,^[21] the Heck^[29] and HWE^[30] reactions, carbonyl group condensations,^[31] and carbonyl-olefin metathesis^[32] represent the methods of choice in this regard.

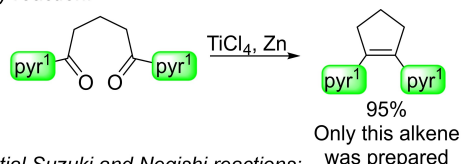
The good availability of acyclic alkenes with two pyrenyl substituents contrasts with the limited availability of cyclic alkenes with two pyrenyl substituents. The latter alkenes can be synthesized via the McMurry reaction, although this procedure has been used to prepare only a single alkene (Scheme 1).^[33] A different approach to the preparation of a cyclic alkene with pyrenyl substituents makes use of the Suzuki and Negishi reactions of bromocyclobutenyl diethyl phosphate.^[34] It is expected that tetrasubstituted cyclic alkenes with pyrenyl moieties may find various applications in the field of materials chemistry. Therefore, based on our experience with cross-coupling reactions,^[35] we decided to develop a new methodology for the preparation of cyclic tetrasubstituted alkenes derived from cyclohexene using the pyrenyl substituent **3** as a model compound for future applications. The proposed approach for the preparation of the cyclohexene **3**

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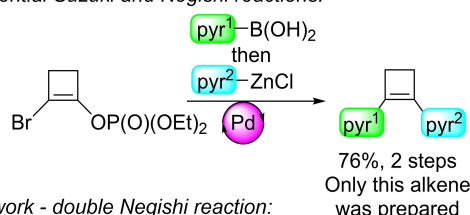
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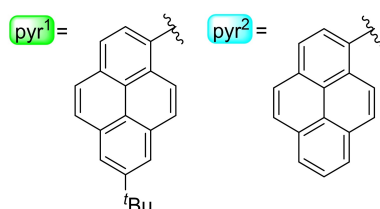
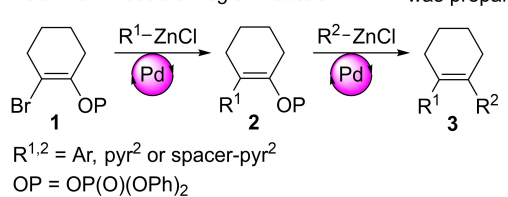
McMurry reaction:



Sequential Suzuki and Negishi reactions:



Our work - double Negishi reaction:

**Scheme 1.** Approaches for the synthesis of cycloalkenes bearing a 1-pyrenyl moiety.

makes use of the regioselective Negishi reaction of a double electrophilic template **1**. Similar double electrophiles are represented by 1-bromo-2-chlorocyclohexene, 1,2-dibromocyclohexene, 2-chlorocyclohexenyl tosylate, and bromocyclohexenyl triflate. Bromochlorocyclohexene can be used in regioselective cross-coupling reactions, but its preparation requires toxic PCl_5 and bromine.^[36] In the case of dibromocyclohexene, the regioselective substitution of two bromine atoms is difficult.^[37] Moreover, halogenated cyclohexenyl tosylates^[38] and triflates^[39] require vigorous reaction conditions to achieve the substitution of both leaving groups.

Results and Discussion

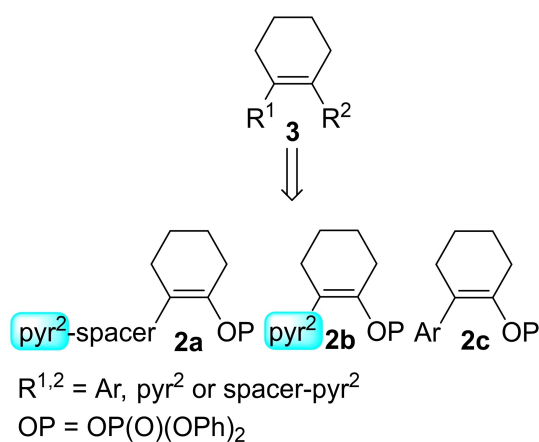
The route proposed for the preparation of cyclic tetrasubstituted alkenes with 1-pyrenyl substituents makes use of the cyclic vinyl phosphate **1** as a model compound for studying the course of the double Negishi reaction to form the alkene **3** (Scheme 1). The choice of cyclic phosphate **1** offers several advantages, including the availability of the starting 2-bromocyclohexanone^[40] and the sufficiently different reactivity of the C–Br and activated C–O bonds for regioselective cross-coupling reactions.^[35c] In addition to the optical properties of the alkene **3**, this work will help to reveal the patterns of the cross-coupling reactions of bromovinyl phosphates with 1-

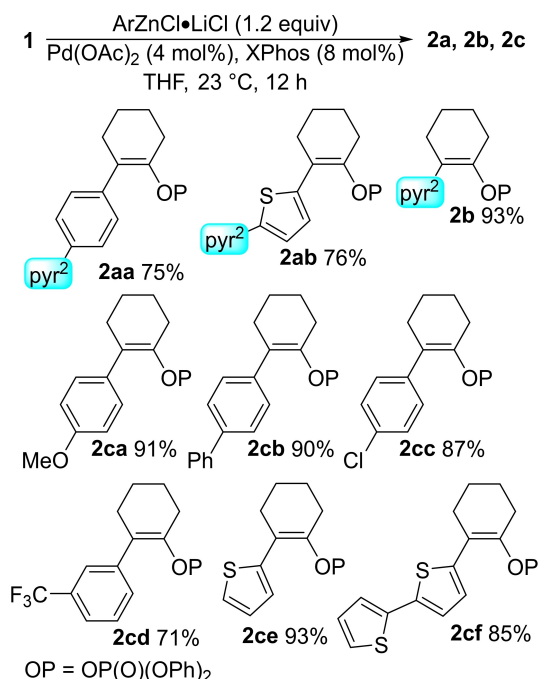
pyrenyl-base organozinc reagents. This is important because these substituents differ in terms of their electronic and steric properties from substituted benzenes, which are more commonly used to determine the extent of cross-coupling reactions.

A simple retrosynthetic analysis showed that three common phosphates, namely **2a**, **2b**, and **2c**, could be designed for the preparation of diverse alkenes with the 1-pyrenyl substituents **3** (Scheme 2). An important intermediate in this case is the phosphate **2b**, the use of which should allow the preparation of the cyclohexene **3b** with two pyrene groups directly attached to the double bond.

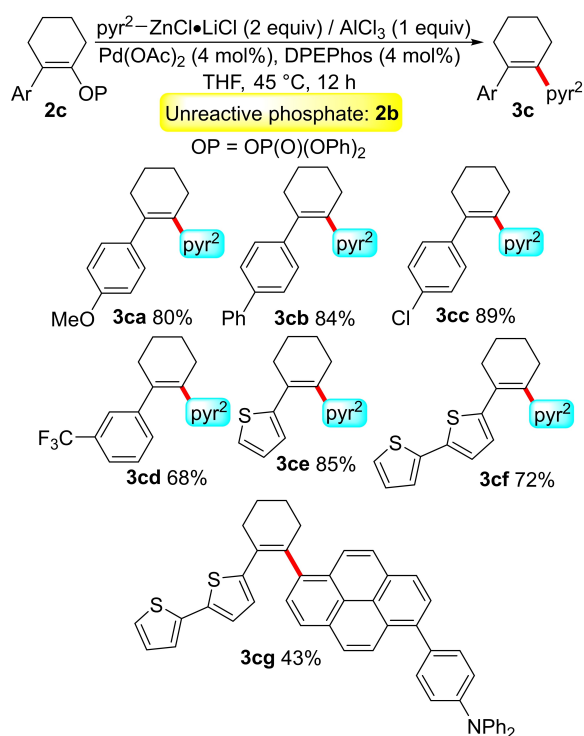
The preparation of the starting phosphates was performed by means of the Negishi reaction of the cyclohexenyl phosphate **1** with organozinc reagents in the presence of both palladium acetate and the dicyclohexyl[2',4',6'-tris(propan-2-yl)[1,1'-biphenyl]-2-yl]phosphane (XPhos) ligand (Scheme 3). The phosphates **2aa**, **2ab**, **2cd**, and **2b** were prepared in good yields. The other phosphates were all obtained in similar isolated yields.^[35c]

Using the prepared phosphates, we studied the scope of the Negishi reaction of the phosphates **2ca–2cf** and **2b** (Scheme 4). In contrast to previous results,^[35c] the tested phosphates **2ca–2cf** smoothly reacted with 1-pyrenylzinc(II) chloride at elevated temperature. The isolated yields of the alkenes **3ca–3cf** ranged from 68–89%. The reaction of the phosphate **2cf** with (6-(4-(*N,N*-diphenylamino)phenyl)pyrene-1-yl)zinc(II) chloride gave the reaction product **3cg** in an average isolated yield. Surprisingly, the phosphate **2b** did not react with 1-pyrenylzinc(II) chloride or 4-methoxyphenylzinc(II) chloride under the tested conditions. Varying the reaction conditions to include an elevated reaction temperature (60 °C and 80 °C in dry THF) and the use of other ligands (DPEPhos, Ruphos, XPhos, etc.) did not prove successful. The substitution of 1-pyrenylzinc(II) chloride for 1-pyrenylmagnesium bromide resulted in the decomposition of the starting phosphate **2b**. We assume that the limited reactivity of the phosphate **2b** during the Negishi reaction was caused by electronic and steric

**Scheme 2.** Retrosynthetic analysis of the synthesis of alkenes with the π -conjugated substituents **3**.

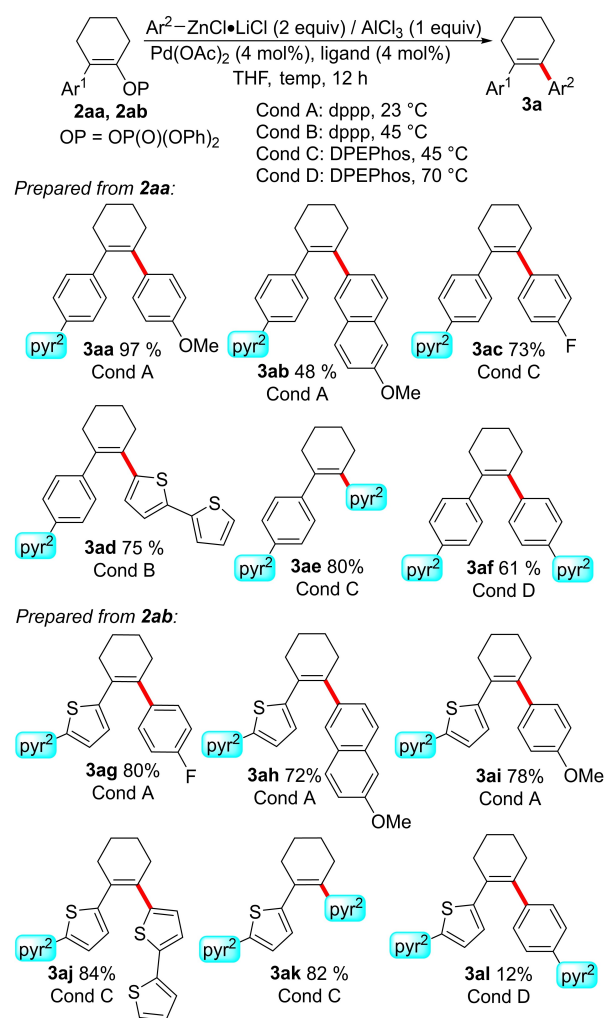


Scheme 3. Synthesis of the starting phosphates.

Scheme 4. Negishi reaction of the phosphate **2c**.

effects of the 1-pyrenyl substituent. However, this assumption requires further verification with respect to the known preparation of bis(pyrenyl)cyclobutene.^[34]

We assumed that the steric or electronic properties of the 1-pyrenyl substituent was responsible for the low reactivity of the phosphate **2b** during the Negishi reaction. Thus, we anticipated that the introduction of a spacer between the double bond and the 1-pyrenyl unit in the phosphates **2aa** and **2ab** would increase their reactivity during the Negishi reaction when compared with the phosphate **2b**. To test this hypothesis, we studied the course of the Negishi reaction of organozinc reagents with the phosphates **2aa** and **2ab**, which both have a 1-pyrenyl moiety installed into the phosphate molecule via a 1,4-phenylene and thiophene-2,5-diyl spacer (Scheme 5). The reaction of the phosphate **2aa** with 4-methoxyphenylzinc (II) chloride gave the corresponding alkene **3aa** in a high yield. The other tested organozinc reagents showed similar reactivity and the alkenes **3ab-3ad** were obtained in satisfactory yields. The same reactivity pattern was observed for the phosphate with a thiophen-2,5-diyl spacer **2ab**. We also successfully carried out the reaction with 1-pyrenylzinc(II) chloride with the phosphates **2aa** and **2ab** to form the dienes **3ae** and **3ak** in high yields. Finally, we took advantage of distinguishing the reactivity of the C–Br and the activated C–O bonds to prepare

Scheme 5. Negishi reactions of the phosphates **2aa** and **2ab**.

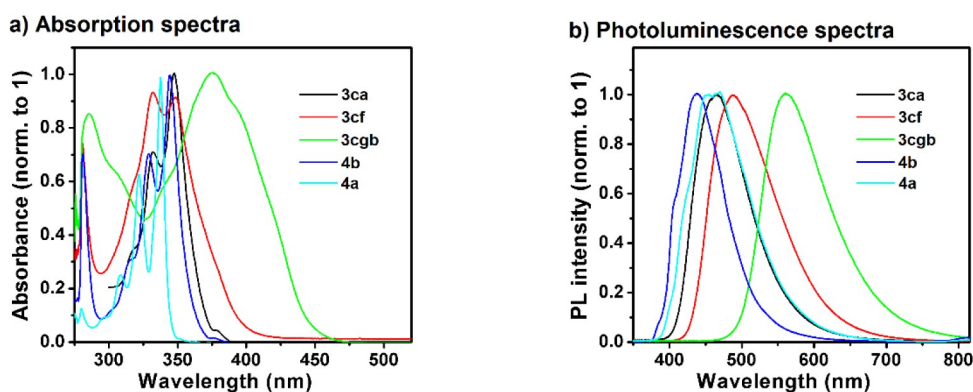


Figure 1. a) Selected normalized absorption spectra of the alkenes **3ca**, **3cf**, **3cgb**, **4a**. Toluene was used as dispersant for the measurements. b) Photoluminescence spectra of the alkenes **3ca**, **3cf**, **3cgb**, **4a**, and **4b**.

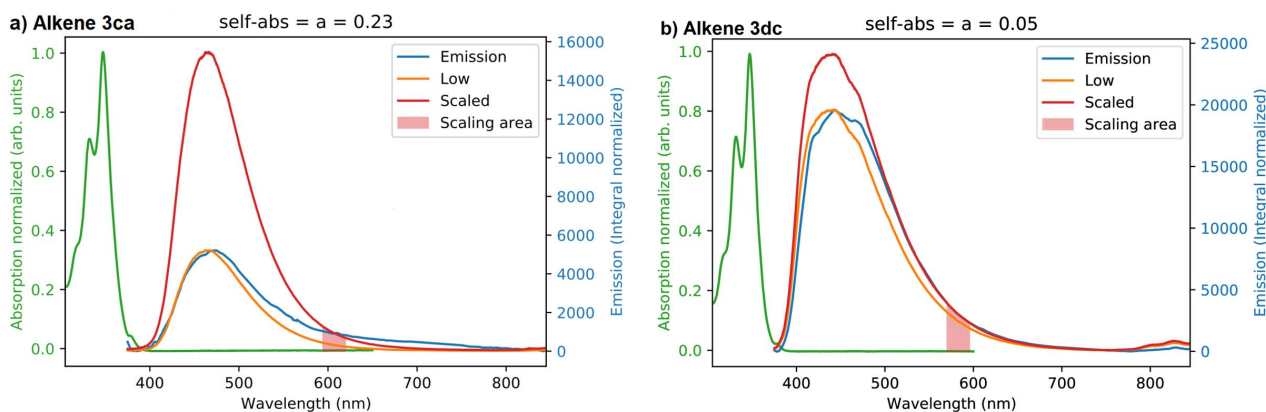


Figure 2. Demonstration of the absorbance and emission spectra of the samples used to determine the self-absorption coefficient a . a) **3ca** with a high self-absorption effect and b) **3dc** with a low self-absorption coefficient.

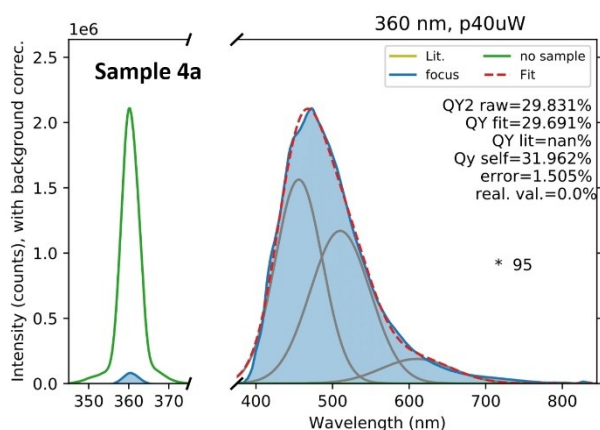


Figure 3. Demonstration of the quantum yield evaluation. The figure contains the spectrum of the excitation without the presence of the samples as well as the spectra of the excitation and emission when the samples were added to the integration sphere.

the cyclic alkenes **3af** and **3al**, which bear 1-pyrenyl substituents connected by two spacers. It is worth noting that the Negishi reactions of phosphates **2aa** and **2ab** for the preparation of alkenes **3aa**, **3ab**, **3ag**, **3ah**, and **3ai** were carried out at room temperature. In other cases, the reaction mixtures had to be heated to 45 °C or 70 °C.

Alkenes with thienyl substituents can be used for the preparation of push-pull molecules (Scheme 6). To demonstrate the post-modification of the prepared alkenes, we selected the alkene **3cg**, which was smoothly lithiated via the reaction with ⁿBuLi. The subsequent reaction with *N,N*-dimethylformamide (DMF) and ethyl chloroformate afforded the disubstituted pyrenes **3cga** and **3cgb** in high isolated yields.

The properties of the synthesized selected alkenes were characterized using standard absorption spectroscopy and steady-state photoluminescence spectroscopy excited at 325 nm. The absorption was measured using a sample in powder form that was dispersed in toluene. The normalized absorbance spectra of selected samples are presented in

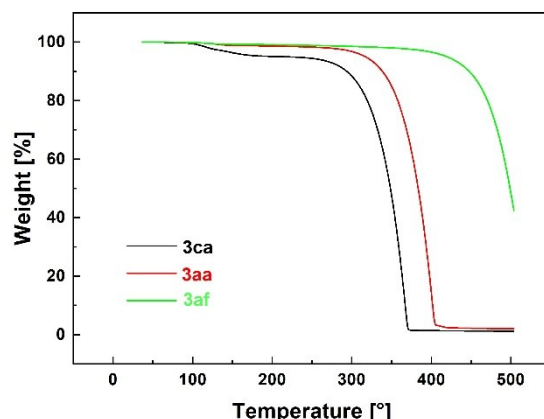
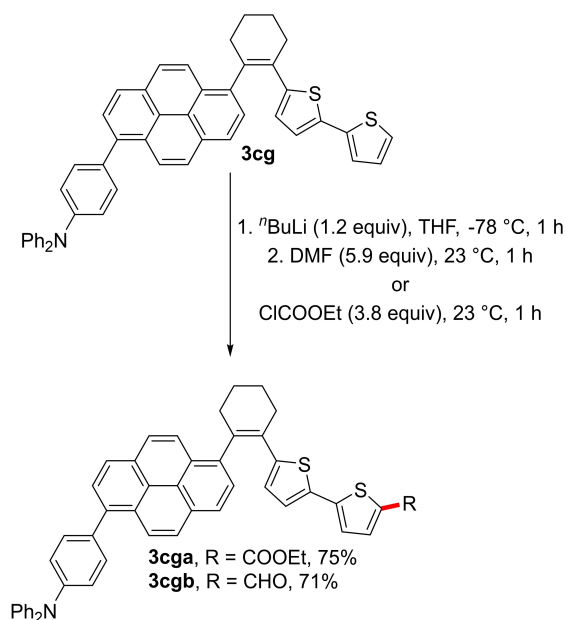


Figure 4. Thermogravimetric analysis thermograms of the alkenes **3ca**, **3aa**, and **3af**.



Scheme 6. Subsequent modification of the phosphate **3cg**.

Figure 1a. The pyrene (**4a**) showed strong absorption maxima at 337 nm and 321 nm (Table 1). Introduction of a cyclohexyl substituent at position 1 of the pyrene ring led to a small bathochromic shift (7 nm) of the alkene **4b**. Further modification of the cyclohexene ring by introducing π -substituents resulted in only a negligible bathochromic shift (≤ 4 nm). This suggests that the pyrene core is deflected from coplanarity, for steric reasons, with π -substituents on the double bond causing limited electronic communication. Unfortunately, we were unable to obtain crystals from the studied alkenes for crystallographic analysis to support this assumption, but the published structure of the 1,2-disubstituted cyclobutenone with a pyrenyl substituent^[35a] supports the proposed hypothesis. A higher bathochromic shift was observed in the case of alkenes **3ai**,

3cg, **3cga** with thienyl substituents. The highest value of bathochromic shift (32 nm) was observed for the push-pull system with formyl and diphenylamino group **3cgb**. Thus, it can be inferred that the significant bathochromic shift of push-pull cyclohexenes correlates with pronounced intramolecular charge-transfer.^[41] However, the study of this phenomenon is the subject of further study.

The photoluminescence (PL) characterizations of the selected alkenes are presented at Figure 1b. Unlike to the absorption, the PL spectra were significantly affected by the presence of substituents. The pyrene (**4a**) and the alkene **4b** have their spectra composed of an intensive band in the blue spectral region.

A hypsochromic shift (29 nm) was observed for the basic alkene **4b** compared to the pyrene (**4a**) (Table 1). A more pronounced hypsochromic shift (65 nm) was observed for alkene **3cd**, but this may be influenced by a significant self-absorption. Alkenes bearing substituted phenyl rings had a low bathochromic shift (≤ 27 nm) as illustrated by selected alkenes **3ca**, **3ce** and **3aa**. On the other hand, a significant bathochromic shift was observed for alkenes **3cg**, **3aj**, **3cga** and **3cgb** with thienyl substituents. A similar situation was also observed for the Stokes shift, which reflects the reorganization energy between the ground and excited state. The typical values of the Stokes shift were in the range of 6000–8800 cm^{-1} , but the alkenes with thienyl substituents **3aj** and **3ad** reached the Stokes shift of 10200 cm^{-1} and 11300 cm^{-1} , respectively.

The measured quantum yields (QYs) of the samples ranged from around 30% (**4a**) to 2.4% (**3cgb**). The QY was calculated using the self-absorption correction (Figures 2, 3). Those samples with broader absorption spectra showed not only a red shift of the spectra, but also a decrease in the QY. This effect was most evident for the sample **3cgb**, which emits around 560 nm, although the QY is approximately 3%. The positions of the PL maxima and the QYs of all the other samples are presented in Table 1. Based on the correlation between the absorption and the PL spectra, the majority of the

Table 1. Comparison of the photoluminescence, absorbance maxima, and quantum yields of selected alkenes.

Alkene	λ_{abs} [nm] ^[a]	λ_{em} [nm] ^[b]	Stokes shift [cm ⁻¹] ^[c]	Φ_{F} [%] ^[d]
3aa	347/280	440	6100	27.1 ± 1.3
3ca	347	465	7300	10.2 ± 0.4
3cb	348/331/280	469	7400	21.6 ± 1.1
3cc	347/280	439	6000	28.2 ± 1.4
3cd	348/280	402 ^[e]	3900	3.0 ± 0.2
3ce	348/281	448	6400	14.6 ± 0.7
3cf	348/332/280	488	8200	11.7 ± 0.6
3cg	367/284	529	8300	9.7 ± 0.5
3ad	345/280	532	10200	7.2 ± 0.3
3af	345/282	489	8300	27.5 ± 1.4
3ai	371/281	487	6400	13.5 ± 0.7
3aj	348/281	573	11300	6.7 ± 0.3
3ak	347/281	497	8700	15.9 ± 0.8
3al	347/281	495	8600	13.2 ± 0.6
3cga	370/285	550	8800	10.3 ± 0.4
3cgb	376/285	561	8800	2.4 ± 0.2
4a	337/322	467	8300	31.9 ± 1.5
4b	344/280	438	6200	19.7 ± 1.0

[a] The spectra were measured on samples prepared by mixing each sample in a powder form with toluene. In case the absorbance spectra contained more significant peaks/bands, all their maxima are presented.

[b] The normalized photoluminescence spectra of the measured samples were excited using a continual laser beam at 325 nm and a power of 100 μ W. All the spectra are corrected on the sensitivity of the detection system. [c] The Stokes shift was calculated between the lowest absorption maxima and the PL one. [d] The quantum yields have been determined by an absolute method using a commercial integrating sphere (IS200-4 from Thorlabs). The magnitude of the quantum yields of all the samples obtained using excitation at 360 nm after the correction of the self-absorption effect. [e] The photoluminescence spectra of the alkene **3cd** were affected by the self-absorption effect, meaning that the presented maxima could not be precise.

samples with broad absorption showed lower QYs and red-shifted PL, particularly the samples **3cgb**, **3cga**, **3aj**, and **3cg**. However, this behavior cannot be generalized, as the sample **3ai** showed broad absorption spectra but blue PL with a QY > 10%, while the sample **3ad** showed visible PL (QY > 7%) with sharp absorption bands and a high Stokes shift. Thus, there does not appear to be a simple correlation between the samples' chemical structures and their optical properties. The overall behavior will most likely be given by a complex interaction.

We also tested the thermal stability of the selected alkenes by means of a thermogravimetric analysis (TG). The results are shown in Figure 4. The thermal decompositions of the alkenes **3aa** and **3ca** started at 130 °C and 96 °C, respectively, and both samples were completely decomposed at around 400 °C (421 °C for **3aa** and 376 °C for **3ca**). Exceptional thermal stability was observed in the case of the alkene **3af**, for which significant decomposition was observed at 358 °C, although the complete decomposition of the studied sample was not observed under the experimental conditions.

Conclusion

We have demonstrated that diphenyl bromocyclohexenyl phosphate is a suitable building block for the preparation of a cyclic tetrasubstituted double bond with π -conjugated substituents. More specifically, the substitution of the bromine and phosphate groups was conducted via the reaction of the starting diphenyl bromocyclohexenyl phosphate with organo-zinc reagents at room temperature in dry tetrahydrofuran. The studied reactions were catalyzed using a palladium catalyst with the XPhos ligand for the substitution of the bromine atom and 1,3-bis(diphenylphosphino)propane or bis[(2-diphenylphosphino)phenyl] ether for the substitution of the phosphate group. The prepared alkenes were studied with regard to their optical properties by means of absorption and photoluminescence spectroscopy. In addition, we observed the high thermal stability of the alkene **3af** through a thermogravimetric analysis. The results obtained suggest that the tetrasubstituted alkenes bearing thienyl and pyrenyl moieties have optical properties with promising applications in materials chemistry. However, further experiments and quantum chemical calculations with respect to: (i) the substitution and *E/Z*-configuration of the double bond, (ii) the influence of heteroatoms of the five-membered ring, and (iii) a position of the pyrene ring are required to tune and explain physical properties of the synthesized alkenes.

Supporting Information Summary

Supporting information includes general information, general procedures and characterization of the prepared compounds (¹H, ¹³C ³¹P NMR and HRMS).

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: Alkenes • C-C coupling • Luminescence • Absorption • Palladium

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